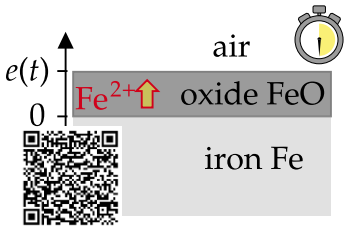
A layer of iron oxide of thickness $e(t)$ grows when hot iron is exposed to air. The oxide layer grows by diffusion of Fe^{2+}. Any Fe^{2+} that reaches the air immediately transforms into solid FeO (s).



Let $D = 10^{-10}$ m^2/s be the diffusion coefficient of Fe^{2+} in FeO (s), $\mu_{Ox} = 5 \times 10^3$kg.m^{-3} the volumetric mass of FeO (s), and $M_{FeO} = 72$ g/mol its molar mass. $n_{Fe^{2+}}(z=0) = n_0 \simeq 10^{28}$ m^{-3}, and Fe^{2+} diffusion is assumed to occur in steady state.

Deduce the duration Δt needed for $e = 1$cm of oxide layer to grow, starting with $e(t=0) = 0$.

ch12.Q1

ANSWER :

On a cross-section S, the increase in mass of the oxide layer between t and $t + \mathrm{d}t$ is :

$$\mathrm{d}M = \mu_{Ox} S \mathrm{d}e = \underbrace{\frac{M_{FeO}}{\mathcal{N}_A}}_{\text{mass of 1 FeO molecule}} \times S j_z \mathrm{d}t$$

Yet, in steady state without volumetric production source, $\Delta n_{Fe^{2+}} = 0$ hence $n_{Fe^{2+}} = n_0 - \frac{z}{e} n_0$, thus, using Fick's law :

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{D n_0 M_{FeO}}{\mu_{Ox} \mathcal{N}_A e} \Rightarrow e(t) = \sqrt{2Kt} \text{ with } K = \frac{D n_0 M_{FeO}}{\mu_{Ox} \mathcal{N}_A}$$

$$\Rightarrow \boxed{\Delta t = \frac{e^2}{2K} = 24 \text{ days}}$$

ch12.A1

Consider the diffusion equation with no source term.

TRUE or FALSE ?

"If the diffusion coefficient D depends on n, the superposition principle no longer holds for solutions of the diffusion equation."

ch12.Q2

ANSWER :

TRUE

Indeed,

$$\frac{\partial n}{\partial t} - D(n)\Delta n = 0 \qquad \text{(a)}$$

is no longer a linear differential equation. For instance, say $D(n) = k \times n$, and suppose that n_1 is a solution of (a)
Yet, $n_2 = 3n_1$ is not solution of (a), because

$$\frac{\partial n_2}{\partial t} - k n_2 \Delta n_2 = 3\frac{\partial n_1}{\partial t} - 9k n_1 \Delta n_1 = 3 \times 0 - 6k n_1 \Delta n_1 \neq 0$$

ch12.A2

TRUE or FALSE ?

"The superposition principle for solutions of the diffusion equation keeps working if there is a non-null α, a constant volumetric production rate."

ch12.Q3

ANSWER :
FALSE

For instance, suppose that n_1 is solution of the diffusion equation, that is : $\frac{\partial n_1}{\partial t} - D\Delta n_1 = \alpha$
Then $n_2 = 2n_1$ should also be solution, yet

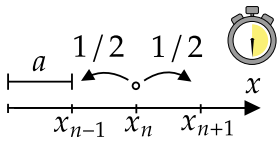
$$\frac{\partial n_2}{\partial t} - D\Delta n_2 = 2\frac{\partial n_1}{\partial t} - 2D\Delta n_1 = 2\alpha \neq \alpha$$

Since $\alpha \neq 0$ here, superposition principle becomes false when there is a production term.

Attention: However, we could superpose a solution n_1 of source term $\alpha_1 \neq 0$ with a solution n_2 of source term $\alpha_2 = 0$, and obtain this way another solution $n_1 + n_2$ for the equation of source term α_1.

ch12.A3

A particle does a 1D random walk. At each time step $t_n = n\tau$, the particle moves according to $x_{n+1} = x_n + \delta x_n$ with :



- $\delta x_n = +a$ with a probability $p = 1/2$
- $\delta x_n = -a$ with a probability $p = 1/2$

Each successive movement δx_i are independent, and $x_0 = 0$. We denote $\langle X \rangle = \mathbb{E}[X]$.

Compute $\langle x_n \rangle$, then $\langle x_n^2 \rangle$, and finally $\sigma_n = \sqrt{\langle x_n^2 \rangle - \langle x_n \rangle^2}$ as a function of t_n.

ch12.Q4

ANSWER :

$$x_n = \sum_{i=0}^{n-1} \delta x_i \Rightarrow \langle x_n \rangle = \langle \sum_{i=0}^{n-1} \delta x_i \rangle = \sum_{i=0}^{n-1} \langle \delta x_i \rangle$$

and $\langle \delta x_i \rangle = \frac{a}{2} - \frac{a}{2} = 0$ hence $\boxed{\langle x_n \rangle = 0}$

$$x_n^2 = \sum_{i=0}^{n-1} \delta x_i \times \sum_{j=0}^{n-1} \delta x_j \Rightarrow x_n^2 = \sum_{i=0}^{n-1} \delta x_i^2 + \sum_{i=0}^{n-1} \sum_{j=0 j \neq i}^{n-1} \delta x_i \delta x_j$$

Yet $\langle \delta x_i \delta x_j \rangle = 0$ if $i \neq j$, since successive movements are independent, and $\delta x_i^2 = a^2$, hence $\boxed{\langle x_n^2 \rangle = na^2}$ and $\boxed{\sigma_n = \sqrt{na^2} = \sqrt{t_n \frac{a^2}{\tau}}}$, with $D = \frac{a^2}{\tau}$.

ch12.A4

In steady state, we observe locally div $\vec{j}<0$.

Provide a physical explanation, explain how this can be achieved and provide examples.

ch12.Q5

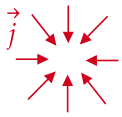
ANSWER :

The conservation equation states

$$\frac{\partial n}{\partial t} + \text{div } \vec{j} = \alpha$$

that is, in steady state div $\vec{j} = \alpha$.

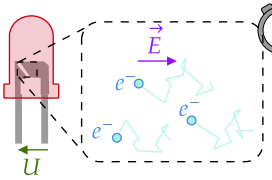
If div $\vec{j} < 0$, then $\alpha < 0$: the volumetric production rate is negative. Therefore, there is a volumetric consumption of the studied chemical.



For instance : O_2 consumed by animal cells, or any chemical reaction consuming the studied chemical.

ch12.A5

In a semi-conductor of constant conductivity γ_0, we study the movement of electrons exposed to a homogeneous electric field $\vec{E}$.



The speed of these conduction electrons is the usual drift speed produced by the $\vec{E}$ field, studied in Ohm's law, superimposed with the diffusion of these electrons . D is their diffusion coefficient, $n(\vec{r},t)$ the number of free electrons by unit of volume, and $-e$ the electric charge of an electron. There is no volumetric production of electrons.

In steady state, determine $\vec{\nabla} n$ using $\vec{E}$, γ_0, e and D.

ch12.Q6

ANSWER :

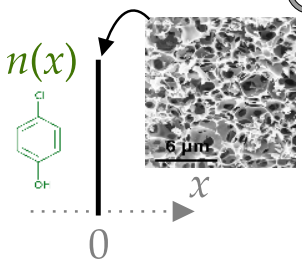
Note $\vec{j_n}$ the flux density vector and $\vec{j}$ the current density vector, such that $\vec{j} = -e\vec{j_n}$.
The movement is the superposition of electrical conduction and diffusion hence the speed of the electrons $\vec{v} = \vec{v}_{\text{Ohm}} + \vec{v}_{\text{diff}}$. Multiplying by $n \Rightarrow \vec{j_n} = \vec{j}_{n,\text{Ohm}} + \vec{j}_{n,\text{diff}}$.
Using Ohm's law $\vec{j}_{\text{Ohm}} = \gamma_0 \vec{E} = -e\,\vec{j}_{n,\text{Ohm}}$ and Fick's law $\vec{j}_{n,\text{diff}} = -D\vec{\nabla} n$ we get the equation of conservation :

$$\frac{\partial n}{\partial t} + \text{div } \vec{j_n} = 0 \underset{\text{steady state}}{\Rightarrow} -\frac{\gamma_0}{e}\vec{E} = D\vec{\nabla} n + \overrightarrow{\text{rot}}\vec{A}$$

with $\vec{A}$ an unknown field, yet $\vec{E} = \vec{0} \Rightarrow \vec{\nabla} n = \vec{0}$ thus $\boxed{\vec{\nabla} n = -\frac{\gamma_0}{eD}\vec{E}}$ *see ex. "biased random walk"*

ch12.A6

A carbon-activated porous layer is used in $x = 0$ to filter 4-chlorophenol from an aqueous solution.
The system is assumed to be in steady state and 1D : the number of 4-chlorophenol molecules per unit of volume n only depends on x.



State the boundary condition at $x = 0$, knowing that the carbon-activated thin layer ensures perfect absorption of 4-chlorophenol.

ch12.Q7

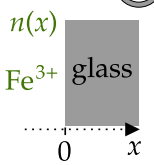
ANSWER :

Imagine N radioactive particles that have a probability $\frac{\mathrm{d}t}{\tau}$ to decay during $\mathrm{d}t$. Then, $\mathrm{d}N = N \times \frac{\mathrm{d}t}{\tau}$ hence $\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{1}{\tau}N$. $\frac{1}{\tau}N$ is the "speed" of the transformation and is obviously proportional to N : the more particles, the more there are particles able to react.
At the wall, the density flux of particles $j_x(0) = -D\frac{\mathrm{d}n}{\mathrm{d}x}$ gets perfectly absorbed that is $-D\frac{\mathrm{d}n}{\mathrm{d}x} = \alpha n(0)$ with $\alpha \propto \frac{1}{\tau}$ and τ the characteristic time for this absorption reaction to occur. Perfect absorption is instantaneous hence $\tau \to 0$ thus $n(0) \to 0$.
Therefore, the concentration at the contact of the layer must vanish : $\boxed{n(0) = 0}$

ch12.A7

A iron ion aqueous solution is in contact at $x = 0$ with the glass of the beaker.

The glass of the beaker is completely impermeable to iron ion and does not chemically interact with them.



The system is assumed to be in steady state and 1D : the number of iron ions per unit of volume n only depends on x.

State the boundary condition at $x = 0$.

ch12.Q8

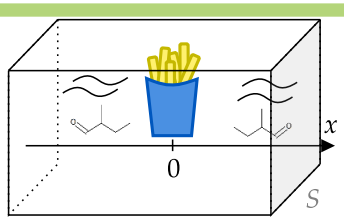
ANSWER :

Since this layer ensures perfect impermeability, no iron ion can pass through. Consequently, the flux $\dot{N}$ of particles through a surface $\mathrm{d}S\vec{e_x}$ equals 0, that is $\vec{j}(x = 0).\vec{e_x}\mathrm{d}S$ hence :

$$\boxed{\vec{j}(x = 0).\vec{e_x} = 0}$$

(and $\frac{\mathrm{d}n}{\mathrm{d}x}(x = 0) = 0$ as well due to Fick's law)

ch12.A8



2-Methylpropanal (2MP), part of the smell of french fries, diffuses in 1D in the air of a long corridor of section S, with a diffusion coefficient D, and $n(0) = n_0$. We neglect convection.

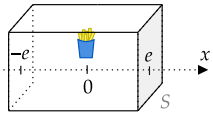
In $x = 0$, a source term $\alpha > 0$ provides $\alpha \mathrm{d}t$ new 2MP molecules to the system between t and $t + \mathrm{d}t$. $\alpha(x \neq 0) = 0$.

Determine the shape of the steady state solution $n(x)$, does it seem acceptable ?

ch12.Q9

ANSWER :

In steady state and 1D, the conservation of 2MB within a right pad of section S and height $2e$ centered over $x = 0$ yields :



$$(j_x(e) - j_x(-e))\, S = \alpha$$

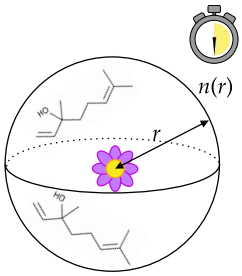
By symmetry of the problem, $n(x)$ should be an even function hence $j_x(x) = -j_x(-x)$ and $j_x(e) = \frac{\alpha}{2S}$ $\forall e \in \mathbb{R}^+$ which yields :

$$\frac{\mathrm{d}n}{\mathrm{d}x}(x) = -\frac{\alpha}{2SD}\frac{x}{|x|} \Rightarrow n(x) = -\frac{\alpha}{2SD}|x| + n_0$$

This solution cannot be acceptable physically since it implies $n(x) < 0$ for $|x| > \frac{2SDn_0}{\alpha}$, and $j_x(0)$ is discontinuous.

ch12.A9

A point-like flower acts as a source of linalool molecules, which then diffuse in the surrounding air with a diffusion coefficient D. In $r = 0$, the source term $\alpha > 0$ provides $\alpha \mathrm{d}t$ new linalool molecules to the air between t and $t + \mathrm{d}t$. $\alpha(r \neq 0) = 0$. The problem is invariant by rotation and $n(r) \underset{r \to +\infty}{\to} 0$.



In steady state, determine $n(r)$ the number of linalool molecules per unit of volume at distance r from the flower.

Hint : $n(r) \Rightarrow \overrightarrow{\mathrm{grad}}\, n = \frac{\mathrm{d}n}{\mathrm{d}r}\vec{e_r}$

ch12.Q10

ANSWER :

$n(r)$ in steady state since the problem is invariant by rotation, hence $\vec{j} = -D\overrightarrow{\mathrm{grad}}\, n = -D\frac{\mathrm{d}n}{\mathrm{d}r}\vec{e_r} = j_r(r)\vec{e_r}$

In steady state, the conservation of linalool within a ball of radius R yields :

$$\mathrm{d}N_{\text{leaving}} = \mathrm{d}N_{\text{appearing}} \Rightarrow 4\pi R^2 j_r(R) = \alpha$$

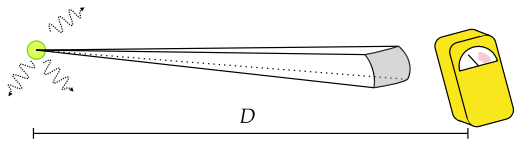
that is :

$$\frac{\mathrm{d}n}{\mathrm{d}r}(R) = -\frac{\alpha}{4\pi D R^2}$$

Therefore $\boxed{n(r) = \frac{\alpha}{4\pi D r} + 0}$

ch12.A10

A geiger counter records 300 hits per minute on its square sensor of size $a \times a$ ($a = 2$cm), when placed at a distance $D = 2$ m of the radioactive source.



The sensor is oriented so that the square sensor has its normal aligned with the source-sensor axis. We suppose that the source emission is isotrope, in steady state, and that any radioactive particle passing through the square sensor is counted as a hit.

Deduce the number of radioactive particles emitted per unit of time by the source.

ch12.Q11

ANSWER :

Note $\mathrm{d}N$ the total number of emitted radioactive particles between t and $t + \mathrm{d}t$.

sensor oriented with its normal aligned with the source-sensor axis $\Rightarrow$ the number of particles passing through the sensor is :

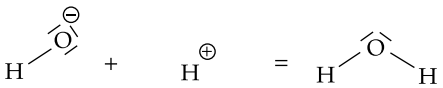
$$\mathrm{d}N_s = \mathrm{d}N \times \frac{\Omega}{4\pi} = \mathrm{d}N\frac{a^2}{4\pi D^2}$$

If there was an angle $\theta \neq 0$ between the sensor's normal and the source-sensor axis, a factor $\cos\theta$ would arise.

Hence :

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{4\pi D^2}{a^2}\frac{\mathrm{d}N_s}{\mathrm{d}t} \Rightarrow \frac{\mathrm{d}N}{\mathrm{d}t} = 3,8 \times 10^7\ \mathrm{s}^{-1}$$

ch12.A11



The chemical reaction $\mathrm{H^+(aq) + HO^-(aq) \to H_2O\ (\ell)}$ occurs at a finite speed.

Therefore, the molar concentration in $\mathrm{H^+}$ varies of $\mathrm{d[H^+]_{reac}}$ between t and $t + \mathrm{d}t$ due to the reaction locally. Van't Hoff law states that $\mathrm{d[H^+]_{reac}} = -k\mathrm{d}t[\mathrm{H^+}][\mathrm{HO^-}]$ with k a constant and [X] the molar concentration of X.

We note $D_{\mathrm{H^+}}$, $D_{\mathrm{HO^-}}$ and $D_{\mathrm{H_2O}}$ the diffusion coefficients.

Write the differential equations that $[\mathrm{H^+}](\vec{r}, t)$, $[\mathrm{HO^-}](\vec{r}, t)$ and $[\mathrm{H_2O}](\vec{r}, t)$ verify.

ch12.Q12

ANSWER :

Attention : $n_{\mathrm{H^+}} = \mathcal{N}_A[\mathrm{H^+}]$. One subsequently gets a production term that is negative for $\mathrm{H^+}$ and $\mathrm{HO^-}$ but positive for $\mathrm{H_2O}$, that yields :

$$\begin{cases} \frac{\partial[\mathrm{H^+}]}{\partial t} = & D_{\mathrm{H^+}}\Delta[\mathrm{H^+}] - k[\mathrm{H^+}][\mathrm{HO^-}] \\ \frac{\partial[\mathrm{HO^-}]}{\partial t} = & D_{\mathrm{HO^-}}\Delta[\mathrm{HO^-}] - k[\mathrm{H^+}][\mathrm{HO^-}] \\ \frac{\partial[\mathrm{H_2O}]}{\partial t} = & D_{\mathrm{H_2O}}\Delta[\mathrm{H_2O}] + k[\mathrm{H^+}][\mathrm{HO^-}] \end{cases}$$

Slightly more advanced example

ch12.A12

In a given metal, we observe a number n of free electrons per unit of volume, all with the same diffusion coefficient D. Each electron has a charge $-e$.

Due to an applied voltage, these free electrons also have an average velocity $\langle \vec{v} \rangle$.

Deduce the density of flux vector $\vec{j}_n$ and the density of electric current vector $\vec{j}$ as a function of the given data.

ch12.Q13

ANSWER :

$\vec{v} = \langle \vec{v} \rangle + \vec{v}_{\text{random}}$ hence, multiplying by n :

$$\vec{j_n} = \vec{j}_{n,\text{drift}} + \vec{j}_{n,\text{diffusion}} \qquad \text{with } \vec{j}_{n,\text{drift}} = n\langle \vec{v} \rangle$$

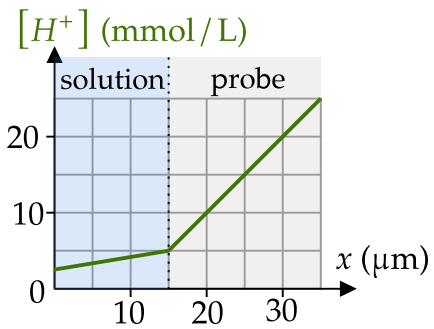
Using Fick's law :

$$\boxed{\vec{j_n} = n\langle \vec{v} \rangle - D\vec{\nabla} n}$$

Since $\vec{j} = -en\vec{v}$ we get also :

$$\boxed{\vec{j} = -en\langle \vec{v} \rangle + eD\vec{\nabla} n}$$

ch12.A13



In steady state, we observe this concentration profile at an interface aqueous solution /probe. The diffusion coefficient in the probe is $D_{\text{H}^+,\text{probe}} = 1 \times 10^{-9}$ m^2.s^{-1}. The membrane is perfectly permeable to the H^+ ions. Deduce $D_{\text{H}^+,\text{solution}}$.

ch12.Q14

ANSWER :

Since the membrane is perfectly permeable to the H^+ ions, $j_x(0^-) = j_x(0^+)$. According to Fick's law :

$$-D_{\text{H}^+,\text{solution}} \frac{\text{d}n_{\text{H}^+}}{\text{d}x}(0^-) = -D_{\text{H}^+,\text{probe}} \frac{\text{d}n_{\text{H}^+}}{\text{d}x}(0^+)$$

$n_{\text{H}^+} = \mathcal{N}_A \times [\text{H}^+]$ hence :

$$D_{\text{H}^+,\text{solution}} = D_{\text{H}^+,\text{probe}} \frac{\frac{\text{d}[\text{H}^+]}{\text{d}x}(0^+)}{\frac{\text{d}[\text{H}^+]}{\text{d}x}(0^-)}$$

We read $\frac{\text{d}[\text{H}^+]}{\text{d}x}(0^+) \simeq 1\text{mmol/L}/\mu\text{m}$ and $\frac{\text{d}[\text{H}^+]}{\text{d}x}(0^-) \simeq \frac{1}{6}\text{mmol/L}/\mu\text{m}$ therefore : $\boxed{D_{\text{H}^+,\text{solution}} \simeq 6 \times 10^{-9}\text{m}^2.\text{s}^{-1}}$

(exceptionally high value due to special interactions between H^+ and H_2O)

ch12.A14

ch12.Q15

ANSWER :

ch12.A15

ch12.Q16

ANSWER :

ch12.A16